

29T CSCA — 2019
COMPUTER SCIENCE AND APPLICATION (Theory)

Full Marks: 70 | Pass Marks: 21 | Time: Three hours

Questions with Model Answers (each question followed immediately by its answer)

Question 1 — Answer the following questions: (1×8=8)

Q: (a) What is a derived class?

A: A derived class is a class that is created by inheriting properties (data members) and behaviour (member functions) from another already-existing class, called the base class. The derived class can also add its own new members or redefine the inherited ones.

Q: (b) Define Queue.

A: A Queue is a linear data structure in which insertion of a new element takes place at one end (called REAR) and deletion of an element takes place at the other end (called FRONT). It works on the FIFO (First In First Out) principle — the element inserted first is removed first.

Q: (c) What is visibility mode?

A: Visibility mode (also called access specifier of inheritance) is the mode — public, private, or protected — mentioned while deriving a class from a base class. It decides how the accessibility of the inherited members of the base class changes in the derived class.

Q: (d) What is a linear data structure?

A: A linear data structure is one in which data elements are arranged in a sequential order, one after another, so that every element (except the first and last) has a unique predecessor and successor. Examples: Array, Stack, Queue, Linked List.

Q: (e) Write the name of any one operation performed on STACK.

A: PUSH — the operation of inserting/adding a new element onto the top of the stack. (POP — the operation of deleting the topmost element — is also a valid answer.)

Q: (f) Define inheritance.

A: Inheritance is an important feature of Object Oriented Programming by which a new class (derived/child class) acquires the properties (data members) and behaviour (member functions) of an already existing class (base/parent class), allowing code reusability.

Q: (g) What is Internet?

A: Internet is a global network that connects millions of computers and computer networks across the world, allowing them to communicate with each other and share information, data and resources using a common set of protocols (TCP/IP).

Q: (h) What does FRONT signify in a queue?

A: FRONT is a pointer/index in a queue that points to the position of the first (oldest) element of the queue — the element that will be deleted next when a DELETE operation is performed.

Question 2 — Answer the following questions: (2×10=20)

Q: (a) When is a copy constructor called?

- When an object is initialized with another object of the same class, e.g. `ClassName obj2 = obj1;` or `ClassName obj2(obj1);`

- When an object is passed by value as an argument to a function.
- When an object is returned by value from a function.

Q: (b) Write two differences between primitive and non-primitive data types.

Primitive Data Type	Non-primitive Data Type
Basic, built-in data types of a language (int, char, float, double).	Derived from primitive data types; user-defined or built using them (array, class, structure, pointer).
Stores only a single value at a time.	Can store a collection of values or references to memory locations.

Q: (c) Differentiate between linear queue and circular queue.

Linear Queue	Circular Queue
Rear end is fixed; once REAR reaches the last index, no more insertion is possible even if front slots are empty.	The last position is connected back to the first position, forming a circle, so empty front slots can be reused.
Leads to wastage of memory space.	Memory space is utilised efficiently; no wastage.

Q: (d) What is POS? Write with example.

A: POS (Product of Sums) is a way of writing a Boolean expression as an AND (product) of one or more OR (sum) terms, where each sum term may contain a variable in either true or complemented form. Example: $Y = (A + B) \cdot (A' + C) \cdot (B + C')$

Q: (e) Draw the logical circuit diagram for DeMorgan's first theorem.

A: DeMorgan's First Theorem states: $(A + B)' = A' \cdot B'$. On the LHS, inputs A and B are first passed through an OR gate, and the output of the OR gate is passed through a NOT gate. On the RHS, A and B are individually passed through separate NOT gates, and the two outputs (A' and B') are then passed through an AND gate. Both circuits give the same output, proving the theorem.

```
LHS: A -- \
      OR-- NOT --> (A+B) '
      B -- /

RHS: A -- NOT -- A' -- \
      AND --> A' . B '
      B -- NOT -- B' -- /
```

Q: (f) Differentiate between LAN and WAN.

LAN (Local Area Network)	WAN (Wide Area Network)
Covers a small geographical area such as a single building, office or campus.	Covers a large geographical area, e.g. a country or the whole globe.
Data transfer speed is very high and error rate is low.	Data transfer speed is comparatively lower and error rate can be higher.
Setup cost is low; usually owned by a single organisation.	Setup cost is high; often uses leased/public communication lines.

Q: (g) What is the role of a switch in a network?

A: A switch is a networking device that connects multiple computers/devices within the same LAN. It receives data packets, reads the destination MAC address, and forwards the packet only to the specific device it is meant for (instead of broadcasting to all devices), which reduces network traffic and collisions and improves efficiency.

Q: (h) Differentiate between static and dynamic allocation of memory.

Static Allocation	Dynamic Allocation
Memory is allocated at compile time.	Memory is allocated at run time (using new/delete in C++).

Size of memory must be known/fixed in advance and cannot be changed during execution.	Size can be decided and changed during program execution, giving flexibility.
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Q: (i) How are files in binary format different from files in text format in C++?

Text Format File	Binary Format File
Data is stored as readable ASCII characters, line by line.	Data is stored exactly as it is held in memory (in binary/machine form).
Can be opened and read directly with a normal text editor.	Cannot be read directly with a text editor; needs a program to interpret it.
Some conversion overhead while reading/writing (e.g. numbers stored as text).	Faster read/write and takes less storage space since no conversion is needed; opened using ios::binary mode.

Q: (j) What is bandwidth? How is it measured?

A: Bandwidth is the amount of data that can be transmitted over a network/communication channel in a fixed unit of time. It represents the data-carrying capacity of the channel. It is measured in bits per second (bps), and commonly in larger units such as kilobits per second (kbps), megabits per second (Mbps) and gigabits per second (Gbps).

Question 3

Q: (a) Rewrite the following program after removing the syntactic error(s), if any: (3)

```
include <iostream.h>
class BUS
{ long Bus No;
  char Description [30];
public
  void Entry ( )
  { cin>>Bus No; gets (Description);
  }
  void Show ( )
  { cout<<Bus No<<":" << Description<<endl;
  }
};
void main ( )
{ Bus B;
  Entry.B ( );
  Show.B ( );
}
```

A: Corrected program:

```
#include<iostream.h>
class BUS
{
long BusNo;
char Description[30];
public:
void Entry()
{
cin>>BusNo;
gets(Description);
}
void Show()
{
cout<<BusNo<<":"<<Description<<endl;
```

```

}
};
void main()
{
    BUS B;
    B.Entry();
    B.Show();
}

```

A: Errors corrected: (1) 'include' changed to '#include'. (2) 'public' needs a colon → 'public:'. (3) Data member/variable name 'Bus No' contained an illegal space — changed to 'BusNo' throughout. (4) The object was declared as 'Bus B;' but the class name is 'BUS' (C++ is case-sensitive) — corrected to 'BUS B;'. (5) Member functions must be called using the syntax object.function(), not function.object() — 'Entry.B();' and 'Show.B();' corrected to 'B.Entry();' and 'B.Show();'.

(b) Answer questions (i) and (ii) after going through the EXAM class code (FUNCTION 1: default constructor, FUNCTION 2: parameterized constructor, FUNCTION 3: ~EXAM(), FUNCTION 4: show()). (1+2=3)

Q: (i) Which concept of Object Oriented programming is illustrated by Function 1 and Function 2 together?

A: FUNCTION 1 and FUNCTION 2 together illustrate the concept of Constructor Overloading (a form of Polymorphism) — the class EXAM has two constructors with the same name but different parameter lists (a default constructor and a parameterized constructor).

Q: (ii) What is Function 3 specifically referred to as? When will Function 3 be invoked?

A: FUNCTION 3 is specifically referred to as a Destructor (~EXAM). A destructor is invoked automatically whenever an object of the class goes out of scope or is explicitly/implicitly destroyed (deallocated) — for example, when the program/function in which the object was created ends.

Q: (c) Write a C++ function to find the sum of diagonal elements from a two-dimensional array of type float. Use the array and its size as parameter with float as its return type. (3)

```

float diagonalSum(float arr[][10], int n)
{
    float sum = 0;
    for(int i = 0; i < n; i++)
        sum = sum + arr[i][i];
    return sum;
}

```

Q: (d) Write an algorithm for the evaluation of POST FIX expression. (3)

- Step 1: Start. Create an empty stack.
- Step 2: Read the postfix expression from left to right, one symbol at a time.
- Step 3: If the symbol is an operand, push it onto the stack.
- Step 4: If the symbol is an operator, pop the top two values from the stack (say val2, then val1).
- Step 5: Evaluate val1 (operator) val2 and push the result back onto the stack.
- Step 6: Repeat steps 3–5 until the entire expression is scanned.
- Step 7: At the end, the single value left in the stack is the final result.
- Step 8: Stop.

— Or —

Q: (d-Or) Write an algorithm for performing insertion in linked queue. (3)

- Step 1: Start. Let FRONT and REAR be pointers to the linked queue.
- Step 2: Create a new node; assign the given value to its DATA part and set its NEXT part to NULL.
- Step 3: If the queue is empty (FRONT = NULL), set FRONT = REAR = new node.

- Step 4: Otherwise, set REAR->NEXT = new node, and then update REAR = new node.
- Step 5: Stop.

Q: (e) Explain the concept of Cartesian Product between two tables, with the help of appropriate example. (1+2=3)

A: The Cartesian Product of two tables is the result obtained by combining every row of the first table with every row of the second table. If Table A has 'm' rows and Table B has 'n' rows, the Cartesian Product will have $m \times n$ rows, and its columns will be the union of the columns of both tables. Example: If Table STUDENT has 3 rows and Table COURSE has 2 rows, then STUDENT CROSS JOIN COURSE (i.e. SELECT * FROM STUDENT, COURSE;) will produce $3 \times 2 = 6$ rows in the result, pairing each student with each course.

— Or —

Q: (e-Or) Explain how SQL UNION operator works? Give example. (3)

A: The SQL UNION operator combines the result sets of two or more SELECT statements into a single result set, and automatically removes duplicate rows. The SELECT statements combined must have the same number of columns, and the corresponding columns must have compatible/matching data types.

```
SELECT ename FROM employee1
UNION
SELECT ename FROM employee2;
```

Q: (f) Write an SQL command to create a table (employee) with attributes: ecode (integer), ename (char(25)), grade (char(2)), gross (decimal). Set ecode as the primary key. Attributes should not allow NULL values. Ensure gross must be greater than 1000. (1+2=3)

```
CREATE TABLE employee
(
  ecode INTEGER PRIMARY KEY,
  ename CHAR(25) NOT NULL,
  grade CHAR(2) NOT NULL,
  gross DECIMAL NOT NULL CHECK (gross > 1000)
);
```

— Or —

Q: (f-Or) Explain UNIQUE, PRIMARY KEY and DEFAULT Constraints with the help of examples.

- **UNIQUE** — ensures that all values in a column (or set of columns) are different from one another; unlike **PRIMARY KEY**, a column with a **UNIQUE** constraint can contain one NULL value. Example: email CHAR(30) **UNIQUE**.
- **PRIMARY KEY** — uniquely identifies each row/record of a table; it cannot contain NULL values and there can be only one primary key per table. Example: ecode **INTEGER PRIMARY KEY**.
- **DEFAULT** — automatically assigns a specified default value to a column when no value is explicitly supplied during insertion. Example: city CHAR(20) **DEFAULT** 'Guwahati'.

Q: (g) Draw the logic circuit diagram for the expression: $Y = ab + b'c + c'a$ (3)

A: The expression is in Sum of Products (SOP) form, so it is implemented using three 2-input AND gates (one for each product term) whose outputs are fed into one 3-input OR gate. NOT gates are used to generate the complements b' and c' .

```
a,b ----> AND1 --> (ab) ----\
b',c ----> AND2 --> (b'c) ----> OR --> Y
c',a ----> AND3 --> (c'a) ----/
```

(NOT gates are used separately to obtain b' from b , and c' from c before feeding AND2 and AND3.)

— Or —

Q: (g-Or) Write and prove the Absorption Law. (3)

A: Absorption Law states: $A + AB = A$ and $A.(A + B) = A$

Proof of $A + AB = A$ (using Boolean algebra rules):

$$\begin{aligned} A + AB &= A.1 + AB && (\text{since } A = A.1) \\ &= A.(1 + B) && (\text{taking } A \text{ common}) \\ &= A.1 && (\text{since } 1 + B = 1) \\ &= A && (\text{Proved}) \end{aligned}$$

Q: (h) Verify the following algebraically: $(A'+B').(A+B) = A'.B + A.B'$ (3)

$$\begin{aligned} \text{LHS} &= (A'+B').(A+B) \\ &= A'.A + A'.B + B'.A + B'.B && (\text{distributive law}) \\ &= 0 + A'.B + A.B' + 0 && (\text{since } A.A' = 0 \text{ and } B.B' = 0) \\ &= A'.B + A.B' && (= \text{RHS, Verified}) \end{aligned}$$

— Or —

Q: (h-Or) Reduce the following Boolean expression using K-Map: $F(P,Q,R,S) = \pi(0,3,5,6,7,11,12,15)$ (3)

A: Since the expression is given in Product of Sums (πm) form, the maxterms 0,3,5,6,7,11,12,15 are plotted as 0's in a 4-variable K-Map (variables P,Q,R,S) and grouped in pairs/quads to obtain the simplified sum terms, which are then multiplied (ANDed) to give the reduced POS expression. On grouping the given maxterms, the simplified expression is obtained as: $F = (P+Q+R).(P'+Q'+S').(P'+R'+S')$ — students should draw the actual 16-cell K-map, mark 0's at the given cell positions and form the largest possible groupings to derive this reduced expression step by step.

Q: (i) Explain any three network topologies for networks. (3)

- Bus Topology — all devices are connected to a single common communication line (the 'bus' or backbone). Data sent by any device travels along the bus and is received by all, but accepted only by the intended recipient. Simple and cheap to install, but a fault in the main cable brings down the whole network.
- Star Topology — all devices are connected individually to a central device (hub/switch). All data passes through the central device. Easy to install and manage, and failure of one node does not affect others, but if the central device fails the whole network stops.
- Ring Topology — each device is connected to exactly two other devices, forming a closed loop/ring, and data travels around the ring in one (or both) direction(s) from node to node until it reaches its destination. Failure of one node can disrupt the whole ring unless a dual-ring is used.

— Or —

Q: (i-Or) Write the differences between GPRS and WLL. (3)

GPRS (General Packet Radio Service)	WLL (Wireless in Local Loop)
A packet-switched mobile data service used on 2G/3G cellular networks to provide internet/data access on mobile phones.	A wireless technology that connects subscribers to the local telephone exchange using radio signals instead of physical copper wires.
Mainly used for mobile data/internet communication.	Mainly used to provide fixed telephone line service wirelessly to homes/offices.
Data rate is comparatively low (tens to hundreds of kbps).	Provides better/consistent voice and moderate data connectivity within a fixed local area.

Question 4

Q: Write a function in C++ to count the number of "He" or "She" words present in a text file "STORY.TXT." Given content: "He is going to market. She is cooking in the kitchen". Output should be: COUNT OF He/She in file is 2 (5)

```

#include<fstream.h>
#include<string.h>
void countHeShe()
{
    ifstream fin("STORY.TXT");
    char word[20];
    int count = 0;
    while(fin>>word)
    {
        if(strcmp(word,"He")==0 || strcmp(word,"She")==0)
            count++;
    }
    cout<<"COUNT OF He/She in file is "<<count;
    fin.close();
}

```

A: The function opens STORY.TXT for reading, extracts one word at a time using the extraction operator (which stops at whitespace), compares each word with "He" and "She", and increments the counter on a match. For the given content this correctly produces the output: COUNT OF He/She in file is 2.

— Or —

Q: (Or) Write a program in C++ to exchange the positions of string stored in an array using array of pointers. (5)

```

#include<iostream.h>
#include<string.h>
void main()
{
    char *name[3] = {"Amit","Bina","Chandra"};
    char *temp;
    cout<<"Before exchange:\n";
    for(int i=0;i<3;i++)
        cout<<name[i]<<endl;

    // exchanging position of first and last string
    temp = name[0];
    name[0] = name[2];
    name[2] = temp;

    cout<<"After exchange:\n";
    for(i=0;i<3;i++)
        cout<<name[i]<<endl;
}

```

Question 5

Q: How does the access of inherited members depend upon their access specifiers and the visibility modes of the base class? Explain. (5)

A: When a derived class inherits from a base class, the access specifiers (private, protected, public) of the base class members combine with the visibility mode (public, protected, private) used in the inheritance to decide the final accessibility of the inherited members in the derived class, as summarised below:

Base class member	Resulting access in derived class
-------------------	-----------------------------------

public	public inheritance → public in derived class protected inheritance → protected in derived class private inheritance → private in derived class
protected	public inheritance → protected in derived class protected inheritance → protected in derived class private inheritance → private in derived class
private	not inherited / not directly accessible in any inheritance mode

A: Private members of the base class are never directly accessible in the derived class regardless of the visibility mode; they can only be accessed through public/protected member functions of the base class. Public inheritance keeps the base class interface mostly intact, protected inheritance restricts public members to protected, and private inheritance makes all inherited members private in the derived class.

— Or —

Q: (Or) What is constructor overloading? Write a program in C++ to show constructor overloading. (5)

A: Constructor overloading means defining more than one constructor in the same class, each having the same name (the class name) but a different parameter list (different number/type of parameters). This allows objects of the class to be initialised in different ways, and the compiler selects the appropriate constructor to call based on the arguments used while creating the object.

```
#include<iostream.h>
class Box
{
int length, breadth;
public:
Box() // Constructor 1 - default
{
length = 0; breadth = 0;
}
Box(int l, int b) // Constructor 2 - parameterized
{
length = l; breadth = b;
}
void show()
{
cout<<"Length="<<length<<" Breadth="<<breadth<<endl;
}
};
void main()
{
Box b1; // calls Constructor 1
Box b2(10,5); // calls Constructor 2
b1.show();
b2.show();
}
```

Question 6

Q: Define a class to represent batsmen in a cricket team. Include the following members — DATA

MEMBERS: FIRST NAME, LAST NAME, RUNS MADE, NUMBER OF FOURS, NUMBER OF SIXES;

MEMBER FUNCTIONS: (i) To assign the initial values, (ii) To update runs made, (iii) To display batsman's information. Make appropriate assumptions about access labels. (5)

```
#include<iostream.h>
#include<string.h>
```

```

class Batsman
{
private:
char firstName[20];
char lastName[20];
int runsMade;
int numFours;
int numSixes;

public:
void assignInitial(char fn[], char ln[]) // (i)
{
strcpy(firstName, fn);
strcpy(lastName, ln);
runsMade = 0;
numFours = 0;
numSixes = 0;
}

void updateRuns(int r, int fours, int sixes) // (ii)
{
runsMade += r;
numFours += fours;
numSixes += sixes;
}

void display() // (iii)
{
cout<<"Name : "<<firstName<<" "<<lastName<<endl;
cout<<"Runs Made : "<<runsMade<<endl;
cout<<"Fours : "<<numFours<<endl;
cout<<"Sixes : "<<numSixes<<endl;
}
};

void main()
{
Batsman p1;
p1.assignInitial("Rohit","Sharma");
p1.updateRuns(45,6,2);
p1.display();
}

```

A: Assumption: all data members are kept private (data hiding), and the three required functions are made public so that they can be accessed from outside the class to assign values, update runs, and display information.

— Or —

Q: (Or) What are static data members of a class? Explain the characteristics of a static data member. Write a program in C++ to support your answer. (5)

A: A static data member of a class is a data member that is declared using the keyword 'static'. Only one copy of a static data member is created and shared by all objects of the class, instead of every object having its own separate copy.

Characteristics of a static data member:

- It is initialised to zero by default when the first object of the class is created.

- It is visible only inside the class, but its lifetime is the entire program (it exists even before any object is created, once it is defined outside the class).
- It must be defined (and optionally initialised) outside the class using the scope resolution operator (::), in addition to its declaration inside the class.
- It is shared by all objects of the class — a change made by one object is reflected for all other objects.
- It can be accessed using the class name directly (ClassName::member) or through any object of the class.

```
#include<iostream.h>
class Counter
{
static int count; // static data member declaration
public:
Counter()
{
count++; // incremented every time an object is created
}
void show()
{
cout<<"Number of objects created so far = "<<count<<endl;
}
};
int Counter::count = 0; // static data member definition/initialisation

void main()
{
Counter c1, c2, c3;
c1.show();
c2.show();
c3.show();
}
```

A: Output: The show() function called through c1, c2 or c3 will print the same shared count value = 3, since 'count' is common to all objects.

— *End of Question Paper with Model Answers* —